An Intraday Implied Volatility Database for B3 Equity Options

Data description and research opportunities

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Data described through 17 September 2026. Collection is ongoing.

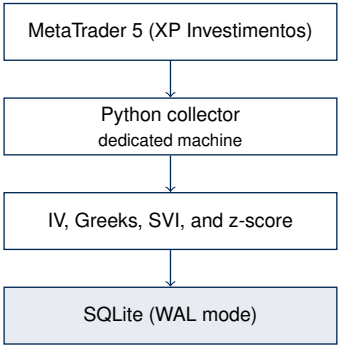
Topics

1. Motivation
2. Data collection and database structure
3. Coverage, descriptive statistics, and signals
4. Data quality and known limitations
5. Research agenda: papers in progress and open questions
6. Next steps

Motivation

- Most studies of Brazilian options rely on end-of-day data, which limits questions about what happens during the trading session.
- The database records the option quotes of seven underlyings throughout the session, with implied volatility, Greeks, and SVI surfaces computed at the time of collection.
- Broad market variables (Ibovespa, US dollar, VIX, and Brent) are stored at the same timestamp, which supports studies of shock transmission.
- This presentation describes the database and organizes potential papers, together with questions that may guide each one. The list is a starting point for discussion within the group.

Data collection and inclusion rules



- Collection window: business days, from 10:00 to 17:10.
- Underlyings: ABEV3, BBAS3, BBDC4, BOVA11, ITUB4, PETR4, and VALE3.
- Moneyness filter: $|k| \leq 0.15$, with $k = \ln(K/F)$.
- Maturity: up to 35 calendar days. One expiration at a time, usually the nearest monthly series, rolled close to expiration.
- Two SVI surfaces per underlying and timestamp, one fitted to calls and one to puts. The `_opp` columns store the surface of the opposite option type.
- The z-score compares current IV with a historical window of 30 to 500 observations of the same series.

Implied volatility: Black-Scholes-Merton

Each option is priced with the Black-Scholes-Merton formula for European options, with continuous rate r and $T = \text{DTE}/252$:

$$c = S_0\, N(d_1) - K\, e^{-rT} N(d_2), \qquad p = K\, e^{-rT} N(-d_2) - S_0\, N(-d_1),$$

$$d_1 = \frac{\ln(S_0/K) + \left(r + \sigma^2/2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

- The implied volatility σ^{imp} is the value of σ that equates the model price to the mid quote, $c(\sigma^{\text{imp}}) = (\text{bid} + \text{ask})/2$, and analogously for puts.
- Delta, gamma, theta, and vega are evaluated at σ^{imp}.
- The database also stores the forward, $F = S_0\, e^{rT}$. Moneyness is measured as log-moneyness relative to it, $k = \ln(K/F)$, and the formula above can be written equivalently on F.
- r is currently fixed at 14.35% per year. Replacing it with the daily DI curve is among the next steps.

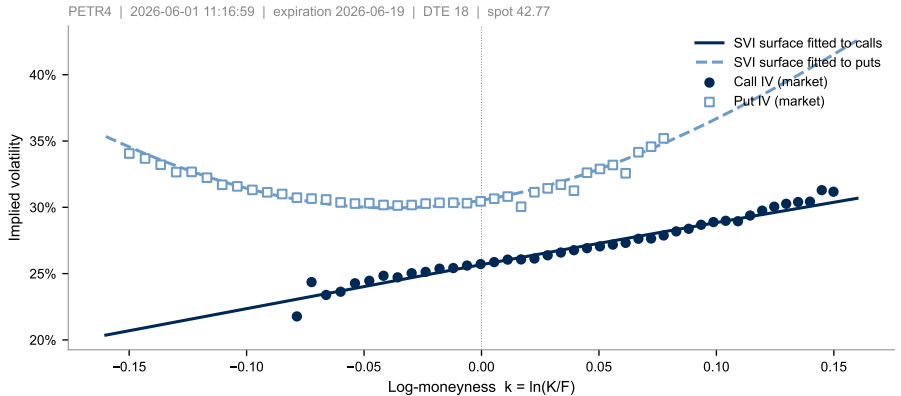
Two SVI surfaces per snapshot: one for calls and one for puts

At each snapshot, the raw SVI parameterization is fitted to total implied variance $w(k) = \sigma^2_{\text{imp}}(k)\,T$:

$$w(k) = a + b\left\{\rho\,(k-m) + \sqrt{(k-m)^2 + \sigma^2}\right\}, \qquad \sigma_{\text{SVI}}(k) = \sqrt{w(k)/T}.$$

- a sets the level, b the angle between the wings, ρ the asymmetry, m the horizontal shift, and σ the curvature near the minimum.
- Calls and puts are fitted separately. The parameters of the surface of each option's own type are stored in `svi_*`, those of the opposite type in `svi_*_opp`, and the fitted value in `iv_svi`.
- With a single joint surface, put-call parity pushes call and put residuals in opposite directions. A signal that requires both legs of a straddle to deviate in the same direction would then be ruled out by construction.
- The z-score is computed from the deviation of each series relative to its own history, and the signal requires both legs to exceed the threshold at the same time.

Call and put smiles in a single snapshot



Markers show market implied volatilities of calls (filled) and puts (hollow). Lines show the SVI surface fitted to each option type at the same timestamp.

The database holds 3.6 million observations over 66 collection days

Table `research_data`	
Period	11 May to 17 Sep 2026
Days with data	66
Observations	3,614,957
Distinct timestamps	106,375
Underlyings	7
Expirations	5
Series (option tickers)	2,294
Distinct strikes	1,156
Variables	47

File `svi_data.db`	
Size	1.6 GB
`research_data`	3,614,957 rows
`iv_history`	810,148 rows
`positions`	10 rows

On complete days, the collector records between 85 and 98 thousand observations and about 1,800 to 2,200 distinct timestamps across the seven underlyings.

`iv_history` feeds the z-score calculation. `positions` stores the positions of the trading system in a demo account.

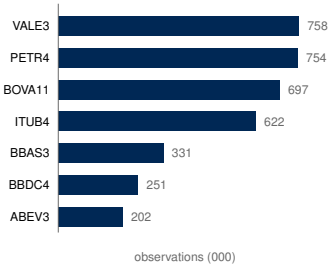
Available variables by group

Group	Variables	Coverage
Identification	`ts, ativo, symbol, tipo, strike, expiration, dte`	100%
Market	`spot, forward, bid, ask, mid, spread_abs, spread_rel, moneyness_k`	100%
	`volume`	99.99%
	`open_interest`	0%
Volatility and Greeks	`iv, delta, gamma, theta, vega`	100%
SVI surface	`svi_a, svi_b, svi_rho, svi_m, svi_sigma, svi_rmse`, `_opp` versions, and `iv_svi`	79.2%
Signal	`z_score, sigma_threshold, n_obs_hist, mean_iv_hist, std_iv_hist`	49.6%
	`signal` (`buy_vol` or `sell_vol`)	2.5%
Broad market	`ibov_spot`	100%
	`usdbrl, vix, oil_brent`	99.6% or more
Interest rate	`risk_free_rate` (constant at 14.35% p.a.)	100%

z-score coverage is lower because it requires a minimum history for each series. The signal is filled only when the z-score exceeds the threshold.

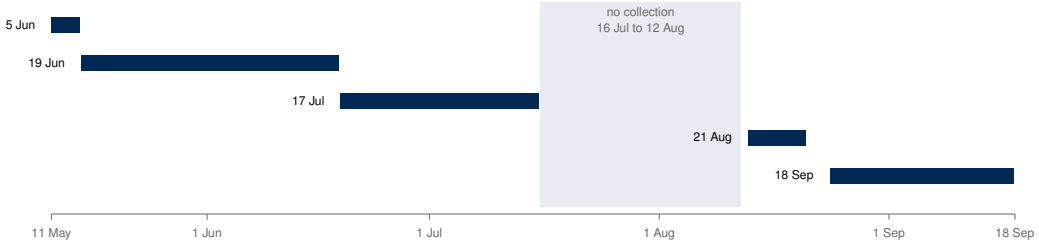
Coverage by underlying

Underlying	Obs. (000)	% total	Strikes	Series	% calls	% SVI
VALE3	758.0	21.0	249	449	64.4	79.0
PETR4	754.2	20.9	230	478	60.7	82.1
BOVA11	696.9	19.3	75	487	62.3	81.9
ITUB4	622.3	17.2	307	366	65.9	76.5
BBAS3	330.5	9.1	180	228	65.9	73.1
BBDC4	251.1	6.9	143	163	65.0	76.7
ABEV3	201.9	5.6	70	123	63.1	81.1



VALE3, PETR4, and BOVA11 account for about 61% of the observations. Calls are the majority for every underlying, between 61% and 66%, reflecting the concentration of liquidity in calls in the Brazilian market.

Expiration cycles: the collector follows one series at a time



Expiration	Obs.	Calls	Puts	DTE	Period covered
5 Jun 2026	78,057	45,602	32,455	22 to 25	11 May to 14 May
19 Jun 2026	1,173,999	632,363	541,636	1 to 35	15 May to 18 Jun
17 Jul 2026	964,763	518,126	446,637	2 to 28	19 Jun to 15 Jul
21 Aug 2026	210,659	118,883	91,776	1 to 8	13 Aug to 20 Aug
18 Sep 2026	1,187,479	983,957	203,522	1 to 25	24 Aug to 17 Sep

Since there is only one expiration per timestamp, the database does not yet allow comparing different maturities at the same moment. The September cycle has a much lower share of puts than earlier cycles.

Selected descriptive statistics

Variable	N (000)	P5	P25	Median	P75	P95
Market IV (`iv`)	3,615	0.149	0.199	0.237	0.281	1.036
Surface IV (`iv_svi`)	2,863	0.146	0.193	0.228	0.263	0.329
Relative spread	3,615	0.003	0.014	0.029	0.074	0.216
Volume (contracts)	3,615	85	200	900	3,000	15,500
DTE (calendar days)	3,615	3	9	16	22	28
SVI fit RMSE	2,863	0.0006	0.0014	0.0034	0.0069	0.0403
z-score	1,792	−1.92	−0.92	−0.13	0.75	2.05
Signal threshold	1,792	1.56	1.68	1.75	1.80	1.99
Ibovespa (points)	3,615	169,630	173,445	176,560	179,650	189,495
VIX	3,602	14.48	15.48	16.12	17.26	19.07
Brent (US$)	3,602	72.13	80.04	90.94	96.24	107.45

- The upper tail of market IV comes from deep out-of-the-money options close to expiration. The surface IV smooths these points.
- The period includes a wide range of Brent prices, from about 70 to 112 dollars, which is useful for shock transmission studies.

Recorded volatility signals

	buy_vol	sell_vol
Observations with signal	35,966	52,722
Days with signal	45	54
Mean z-score	−2.28	2.86
Extreme z-score	−13.67	21.38
Mean threshold	1.69	1.77
Signals below threshold	0	0

Underlying	buy_vol	sell_vol
VALE3	7,206	6,504
PETR4	3,978	7,283
ITUB4	3,575	5,083
BBAS3	1,225	1,621
ABEV3	897	2,186
BOVA11	597	2,314
BBDC4	505	1,370

Counts per leg (call or put).

- Signals always come in pairs, with the same number of calls and puts, consistent with straddle construction.
- `sell_vol` occurs with positive z-scores and `buy_vol` with negative z-scores. No signal has a z-score below the threshold.
- The threshold ranges from 1.51 to 2.54 and is calibrated by series.
- Short volatility signals are more frequent for almost every underlying. VALE3 is the exception.

Data quality and known limitations

Issue	Magnitude	Possible treatment
Business days without collection	27 days, 20 of them between 16 Jul and 12 Aug	Document the gap and handle it in tests that require continuity
Partial days	8 days with few observations or missing underlyings	Exclude or flag in daily analyses
Empty `open_interest`	100% missing	Add to collection. Needed for aggregate gamma and pinning
Constant risk-free rate	14.35% p.a. over the whole period	Store the daily DI curve
Dollar in futures contract points	median of 5,154.5	Divide by 1,000 before use
Crossed quotes (bid $>$ ask)	5,937 rows (0.16%)	Filter before computing mid and IV
IV at solver cap	31 rows	Filter
Non-positive `iv_svi`	25,846 rows (0.72%)	Filter and review the calibration at these timestamps
SVI parameters at bounds	53% of rows	Review bounds, especially $\rho \leq -0.05$
Repeated keys with different tickers	279,700 extra rows	Likely coexistence of weekly, monthly, or adjusted series. Define a selection rule
No data on expiration day	minimum DTE equal to 1	Extend collection to expiration day

There are no exact duplicates on `ts` and `symbol`. The count of SVI parameters at bounds is row-weighted, since parameters repeat within each slice.

Research agenda

Papers in progress and open questions

Where does Black-Scholes-Merton fail? What the database can measure

The implied volatility surface is the map of the pricing error of Black-Scholes-Merton. The research question is how to decompose that error into its sources.

Source of deviation	How the database measures it	Current limitation
Dynamics of the underlying: stochastic volatility, leverage, jumps	Delta-hedged P&L attribution into gamma, vanna, and volga terms from minute-level spot and IV. Skew term structure between 1 and 35 days to expiry.	One expiration per timestamp mixes maturity with market regime
Risk premia	Gap between implied variance and subsequently realized variance, by underlying, moneyness, and maturity.	Four months of data. Evidence is descriptive until the sample grows
Frictions and demand	Divergence between call and put IVs at the same strike, which put-call parity rules out under BSM. Relation with spreads and volume.	No open interest and no trade side
Contract features and implementation	Early exercise of American calls around dividends and interest on equity. Time convention, interest rate, and quote filters.	Must be cleaned first, or it is wrongly attributed to the market

References: Carr and Wu (2016, JFE; 2020, JF) on P&L attribution; Bandi, Fusari, and Renò (2023) on ultra short-tenor options; El Amrani and Guyon on the short-maturity skew.

Black-Scholes-Merton as an accounting tool: delta-hedged P&L attribution

For an option hedged with the BSM delta and financed at rate r, a second-order expansion gives

$$d\Pi \approx \underbrace{\tfrac{1}{2}\,\Gamma S^2\left[\left(\tfrac{dS}{S}\right)^2 - \sigma^2\,dt\right]}_{\text{gamma}} + \underbrace{\mathcal{V}\,d\sigma}_{\text{vega}} + \underbrace{\text{Vanna}\;dS\,d\sigma}_{\text{vanna}} + \underbrace{\tfrac{1}{2}\,\text{Volga}\,(d\sigma)^2}_{\text{volga}},$$

with $d\Pi = dc - \Delta\,dS - r\,(c - \Delta S)\,dt$ and σ the option's own implied volatility.

- If BSM were correct, only the gamma term would remain, with zero expected value. Each additional term corresponds to a violated assumption.
- Gamma: realized versus implied variance, which carries the volatility risk premium and jumps.
- Vega: implied volatility is not constant. Vanna: covariance between the underlying and volatility, related to leverage and skew. Volga: volatility of volatility, related to the curvature of the smile.
- The database stores S, σ, delta, gamma, and vega for every option at each snapshot. Vanna and volga follow in closed form, so each term can be measured contract by contract at minute frequency.

Papers in progress

No.	Title	Authors	Status
2	*SVI Surface Calibration with Asymmetric Liquidity Weights: A Two-Surface Architecture for Emerging Market Options*	Almeida and Farias	**In progress**
5	*Skewed Signals: Options Market Reactions to Green Bond Issuances in an Emerging Market*	Farias	**In progress**
6	*The Volatility Risk Premium in Brazilian Equity Options: Intraday Evidence from a Real-Time Database*	Santos	**In progress**
9	*Intraday Implied Volatility Dynamics and Straddle Profitability in the Brazilian Equity Options Market*	Almeida, Ribeiro, and Farias	**Working paper**
15	*The Greeks Across the Smile: How Delta, Gamma, and Vega Vary with Moneyness and Time to Expiry on B3*	Ávila	**In progress**
17	*Sectoral Heterogeneity in Implied Volatility Dynamics: Banks, Commodities, and the Ibovespa ETF*	Carvalho	**In progress**
18	*Iron Ore Prices and the VALE3 Volatility Smile: An Intraday Analysis of Commodity Spillovers into Options Pricing*	Sena	**In progress**
19	*Oil Price Shocks and the PETR4 Volatility Surface: Real-Time Evidence from Brazilian Equity Options*	Souza	**In progress**
20	*Tax-Exempt Credit: Does Implied Volatility Price Corporate Debt When Fiscal Subsidies Bind?*	Almeida	**In progress**

Numbering follows a single list of topics (1 to 24). The remaining topics are open.

Implied volatility and the volatility surface

1	*Intraday Dynamics of the Implied Volatility Smile in Brazilian Equity Options: Evidence from One-Minute Frequency Data* How do the level, slope, and curvature of the smile behave during the trading session?	**Open**
2	*SVI Surface Calibration with Asymmetric Liquidity Weights: A Two-Surface Architecture for Emerging Market Options* Do liquidity weights and separate fits for calls and puts improve fit quality and parameter stability?	**In progress**
3	*Term Structure of Implied Volatility in Brazil: How Does the Smile Evolve from 30 to 5 Days to Expiry?* How does the shape of the smile change as expiration approaches?	**Open**
4	*Put-Call Symmetry and the Volatility Skew in Brazilian Equity Options: A Microstructure Perspective* Do the call and put surfaces diverge? Is the divergence related to spreads and liquidity?	**Open**
5	*Skewed Signals: Options Market Reactions to Green Bond Issuances in an Emerging Market* Do green bond issuances change the slope of the issuer's volatility surface?	**In progress**

Volatility risk premium

6	*The Volatility Risk Premium in Brazilian Equity Options: Intraday Evidence from a Real-Time Database* What are the size and sign of the volatility risk premium in Brazilian options when measured with intraday data? Realized volatility can be built from the spot prices stored in the database.	**In progress**
7	*Implied vs Realized Volatility in Brazil: Does the VRP Vary Across Underlyings and Market Regimes?* Does the premium differ across underlyings and change with the external environment, measured by the VIX, the dollar, or Copom decisions?	**Open**
8	*Commodity Exposure and the Volatility Risk Premium: Comparing PETR4, VALE3, and Financial Sector Options* Is commodity exposure associated with a premium different from that of banks? Brent is in the database. Iron ore would need to be added.	**Open**

Trading strategies and market microstructure

9 *Intraday Implied Volatility Dynamics and Straddle Profitability in the Brazilian Equity Options Market* — **Working paper**

Do deviations of implied volatility from its historical mean predict straddle returns after costs? Do the results depend on a few events, such as Copom meetings?

10 *Bid-Ask Spreads and Price Discovery in Brazilian Equity Options: A One-Minute Frequency Analysis* — **Open**

Does the options market lead the underlying at high frequency, or the other way around? How do spreads behave around these moves?

11 *Intraday Liquidity Patterns in Brazilian Equity Options: Time-of-Day Effects and Expiry Cycle Dynamics* — **Open**

How do spreads and volume vary during the day and over the expiration cycle?

12 *Market Making and the Cost of Volatility Insurance: Evidence from Option Spread Dynamics on B3* — **Open**

What share of the cost of option-based protection is the spread? How does this share vary with moneyness and maturity?

Greeks and aggregate exposure

13 *Aggregate Dealer Gamma and Intraday Price Pressure in Brazilian Equities: Evidence from Options Market Data* — **Open**

Does aggregate dealer gamma exposure help explain the intraday volatility of the underlying? Requires open interest, currently missing from the database.

14 *Vega-Weighted Net Volatility Demand and Return Predictability in the Brazilian Options Market* — **Open**

Is vega-weighted net demand for volatility related to future returns? The recorded volume does not identify the trade side.

15 *The Greeks Across the Smile: How Delta, Gamma, and Vega Vary with Moneyness and Time to Expiry on B3* — **In progress**

How are observed Greeks distributed across the smile and across maturities? Do they differ from Black-Scholes-Merton Greeks computed with a single volatility?

16 *Stock Pinning and Implied Volatility Near Option Expiration: Evidence from B3* — **Open**

When the underlying approaches a strike with large open interest, does the implied volatility of options at that strike fall in the last days before expiration? Does the market anticipate the lower realized volatility associated with pinning? Requires open interest, collection on expiration day, and more cycles.

Comparisons across underlyings, sectors, and markets

17 *Sectoral Heterogeneity in Implied Volatility Dynamics: Banks, Commodities, and the Ibovespa ETF* — **In progress**

Do implied volatility dynamics differ across banks, commodity firms, and the Ibovespa ETF? Do the differences persist after controlling for liquidity?

18 *Iron Ore Prices and the VALE3 Volatility Smile: An Intraday Analysis of Commodity Spillovers into Options Pricing* — **In progress**

Do iron ore price changes transmit to the VALE3 smile within the day? Requires an intraday iron ore series.

19 *Oil Price Shocks and the PETR4 Volatility Surface: Real-Time Evidence from Brazilian Equity Options* — **In progress**

How fast and in what form do Brent shocks reach the level, slope, and curvature of the PETR4 surface? Is the response asymmetric between price increases and decreases?

20 *Tax-Exempt Credit: Does Implied Volatility Price Corporate Debt When Fiscal Subsidies Bind?* — **In progress**

Does the implied volatility of the issuer's stock help explain spreads on tax-exempt infrastructure debentures? Few issuers are covered by the database.

Methodology and systems

21 *A Real-Time Implied Volatility Database for Brazilian Equity Options: Architecture, Quality Filters, and Descriptive Statistics* — **Open**

How is the database built, which quality filters are applied, and what does it cover? It can serve as a common reference for the group's other papers.

22 *Automatic Threshold Calibration for Volatility Trading Signals: A Coefficient-of-Variation Approach* — **Open**

Does a series-specific threshold, calibrated from the historical dispersion of implied volatility, produce more stable signals than a fixed threshold? The threshold already ranges from 1.51 to 2.54 in the database.

23 *From Signal to Execution: Transaction Costs and Slippage in Systematic Straddle Strategies on B3* — **Open**

How much of the theoretical signal performance is lost in execution? The demo account positions table is still too small for this purpose.

24 *Where Does Black-Scholes-Merton Fail? Delta-Hedged P&L Attribution in Brazilian Equity Options* — **Open**

How much of the pricing error comes from realized variance, spot-volatility covariance, and volatility of volatility? Does the decomposition differ across underlyings, moneyness, and days to expiry?

What the database supports today and what requires adjustments

Requirement	Papers affected
Nothing beyond the current database	1, 2, 4, 6, 9, 10, 11, 12, 15, 17, 19, 21, 22, 24
Longer collection period	5, 9, 16, and 23 (more events), 7 (regimes)
Open interest	13, 16
Trade side classification	14
More than one expiration per timestamp	3 (comparison across maturities)
External series	8 and 18 (iron ore), 20 (debenture spreads)
Collection on expiration day	11 (end of cycle), 16

Indicative classification, subject to discussion with the authors of each paper.

Next steps

Data collection

- Add open interest and the daily DI curve.
- Extend collection to expiration day.
- Evaluate collecting two expirations simultaneously.
- Monitor missing or partial days with automatic alerts.
- Define which series to keep when two tickers share the same strike and expiration.
- Convert the dollar from futures contract points to reais per dollar.

Research

- Use the database description paper (21) as a common reference.
- Freeze dated versions of the database for each paper, which helps replication.
- Share quality filters and data loading code across papers.
- Assign open topics to interested members.